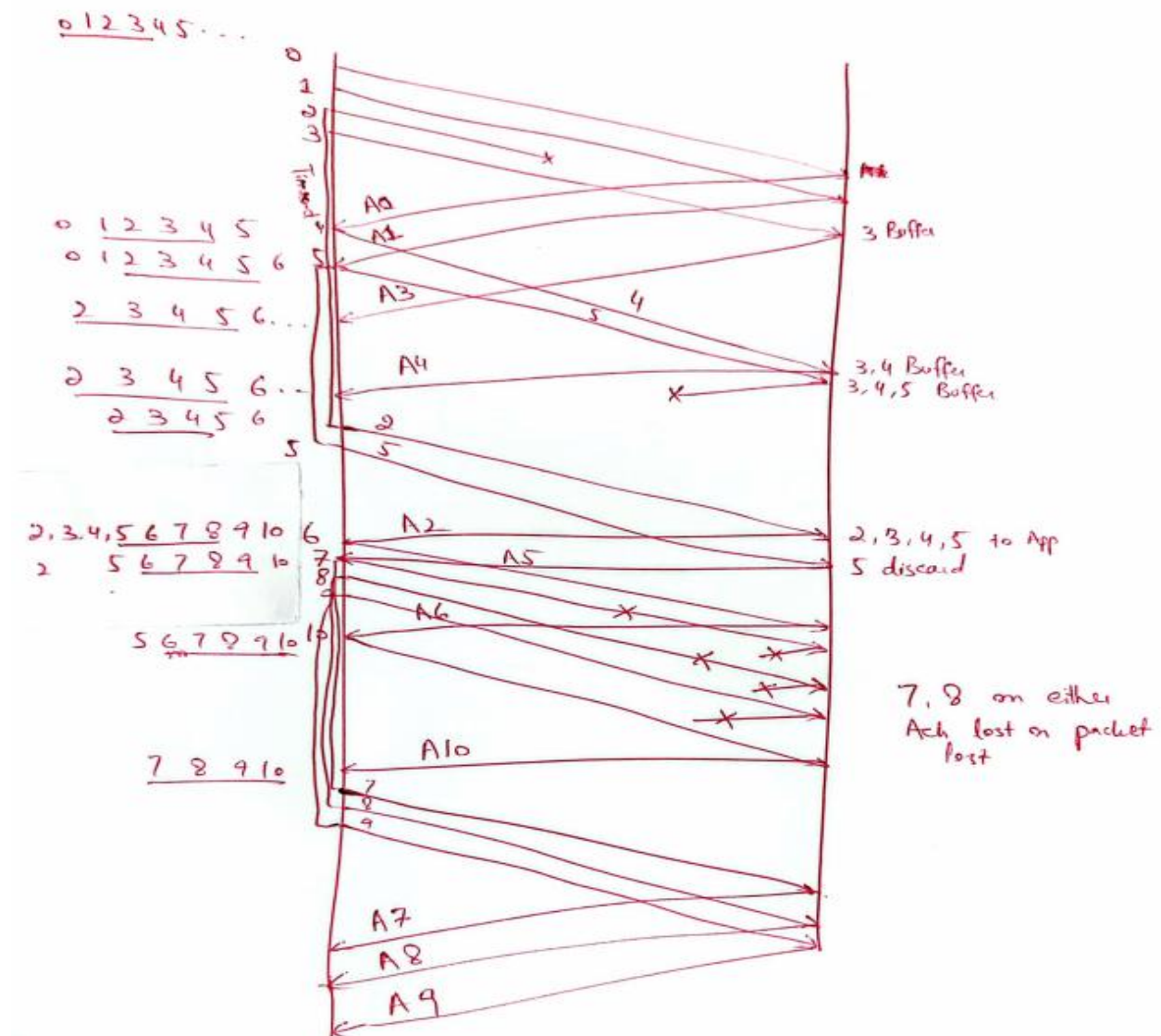


Question 1 [Marks 10 +10]

a) Consider a data transmission scenario between a sender and a receiver using the Selective Repeat protocol. Illustrate the exchange of packets and acknowledgments between the sender and receiver for packets numbered 0 to 10, considering the following events with window size 4:

1. Packet 2 is lost during the first transmission.
2. ACK for packet 5 is lost.
3. Timeouts occur at the sender after packets 7 and 8.
4. Packet 9 ack lost during transmission.

Show the sequence of transmissions, losses, timeouts, and acknowledgments on a timing diagram.



b) Draw sender finite state machines (FSMs) for a protocol RDT-3.1.

1. The channel may corrupt, lose, or delay packets and ACKs.

2. Packets can have sequence number 0,1 and 2. [0,1,2,0,1,2]

3. The sender:

a. Sends one packet at a time and starts a timer after each transmission.

b. Retransmits the packet on timeout or on receiving a duplicate ACK.

c. Ignores garbled ACKs.

4. No NAKs are used.

Q2. Two IoT monitoring devices, Device A and Device B, send sensor data to a cloud analytics server over TCP. **(8 Marks)**

Both devices use the same retransmission timer parameters:

EstimatedRTT = 200 ms, DevRTT = 20 ms, $\alpha = 0.125$, $\beta = 0.25$.

- Device A operates over a stable wired link where RTTs stay close to 200 ms.
- Device B operates over a variable wireless link where RTTs fluctuate widely between 100 ms and 500 ms.

Tasks:

a) Describe how **DevRTT** and **TimeoutInterval** evolve for each device. (2 marks)

For sender A, a) DevRTT decreases overtime. b) TimeoutInterval starts from 280 and move towards 200ms over time.

For sender B, a) DevRTT increases and stays elevated b) timeout **stays large and more volatile**, jumping up after spikes.

b) Without calculation, which device will **retransmit more often**, and why? (1 marks)

Sender B will **retransmit more often** because its RTTs are highly variable: occasional spikes can **exceed the current TimeoutInterval**, triggering timeouts and retransmissions.

c) Suppose Device B doubles β to 0.5. What happens to its **TimeoutInterval** behavior? (1 marks)

More responsive timeout, it expands quickly during spikes and contracts quickly afterward.

d) At $t = 5000$ ms, Device B has EstimatedRTT = 200 ms and DevRTT = 20 ms ($\alpha = 0.125$, $\beta = 0.25$). It sends a segment S4, whose ACK arrives at $t = 5300$ ms. Immediately after that, it sends S5, which is lost and retransmitted. When the ACK for S5 is finally received, Device B sends S6. **Calculate the TimeoutInterval for S6.** (4 marks)

$$\begin{aligned}\text{SampleRTT} &= 5300 - 5000 = 300 \text{ ms} \\ \text{EstimatedRTT} &= (1 - \alpha) \times 200 + \alpha \times 300 = 212.5 \text{ ms} \\ \text{DevRTT} &= (1 - \beta) \times 20 + \beta \times |300 - 212.5| = 36.875 \text{ ms} \\ \text{TimeoutInterval} &= \text{EstimatedRTT} + 4 \times \text{DevRTT} = 212.5 + 4 \times 36.875 = 360 \text{ ms}\end{aligned}$$

Q3. Host A (receiver) has a receive buffer (RcvBuffer) of 4000 bytes. **(7 Marks)**
 Its application reads data at a constant rate of 1000 bytes per second.
 Host B (sender) transmits 1000-byte TCP segments at a rate of 2 segments per second.

Assume:

- The connection is established.
- All segments arrive successfully and in order.
- No retransmissions and delays are ignored.
- The sender stops sending when $\text{rwnd} = 0$.

Tasks:

a. After 4 seconds, how much data is stored in Host A's buffer? **(2 Marks)**

$$\begin{aligned}\text{Data sent} &= 2 \times 1000 \times 4 = 8000 \text{ B} \\ \text{Data read} &= 1000 \times 4 = 4000 \text{ B} \\ \text{Buffered} &= 8000 - 4000 = 4000 \text{ B}\end{aligned}$$

Or 3000 B

b. What is the advertised rwnd at that moment? **(1 Mark)**

0 Or 1000B

c. After another 1 second ($t = 5 \text{ s}$), what is the new rwnd . **(1 Mark)**

- At $t = 5 \text{ s}$, app reads 1000 B $\rightarrow$ buffer = 3000 B; $\text{rwnd} = 1000 \text{ B}$ (The sender stops sending when $\text{rwnd} = 0$.) or in other case, 1000B too.

The first segment starts with sequence number 10 000.

d. What is the sequence-number range of the segment sent at $t = 3 \text{ s}$? **(2 Marks)**

Each second: 2 segments
 so at $t = 3 \text{ s}$, 6 segments have been sent
 Start seq = $10\,000 + (6 \times 1000) = 16\,000$ (range 16 000–16 999).

e. What ACK number will Host A send immediately after receiving it, assuming all prior data arrived correctly? **(1 Mark)**

17000

Question 4 [15 Marks]

A TCP sender follows Reno-style congestion control. The round-trip time (RTT) is 100 ms, and the maximum segment size (MSS) is 1500 bytes. Losses are detected only by triple-duplicate ACKs unless otherwise stated.

- a) Give the average sending rate (in bytes/second) over a Reno cycle in terms of W, RTT, and MSS. [5 Marks]

Average rate over Reno cycle

$$\text{Average cwnd} = (W/2 + W) / 2 = 3W/4 \text{ MSS}$$

$$\text{Average rate} = (3W/4 \times \text{MSS}) / \text{RTT bytes/s}$$

- b) For W = 200 MSS, compute the average sending rate (in Mbps). [3 Marks]

$$\text{Average rate} = (3 \times 200 / 4 \times 1500) / 0.1 = 2,250,000 \text{ bytes/s} = 18.0 \text{ Mbps}$$

- c) Now assume a timeout happens when cwnd reaches 200 MSS. After the timeout:

- cwnd resets to 1 MSS,
- ssthresh is set to 100 MSS,
- cwnd grows exponentially (slow start) until it reaches 100 MSS,
- then cwnd increases linearly by +1 MSS per RTT (congestion avoidance) from 100 MSS up to 200 MSS before the next loss.

- i) How many RTTs are spent in slow start [2 Marks]?

- ii) How many RTTs are spent in congestion avoidance [2 Marks]?

- iii) What is the average sending rate (in Mbps) over the full cycle [3 Marks]?

(c) Timeout resets cwnd = 1 MSS; ssthresh = 100 MSS.

Slow start phase: sequence = 1, 2, 4, 8, 16, 32, 64 → 7 RTTs to reach/exceed 100.

Bytes in SS = $(1 + 2 + 4 + 8 + 16 + 32 + 64) \times \text{MSS} = 127 \text{ MSS} = 190,500 \text{ bytes}$; time = 0.7 s.

Congestion avoidance phase: from 100 MSS to 200 MSS (+1 per RTT) = 100 RTTs.

Bytes in CA = $(100 + 199)/2 \times 100 \text{ MSS} = 14,950 \text{ MSS} = 22,425,000 \text{ bytes}$; time = 10 s.

Overall: total bytes = 22,615,500; total time = 10.7 s → average = $22,615,500 / 10.7 = 2,113,646 \text{ bytes/s} \approx 16.9 \text{ Mbps}$.

- (i) 7 RTTs, (ii) 100 RTTs, (iii) 16.9 Mbps average rate.

Question 5 [10 Marks]

Two long-lived flows share a bottleneck link with capacity $R = 80$ Mbps. Due to losses and timeouts, each flow's transport adds 20% retransmission overhead beyond its original data rate: $\lambda_{in'} = 1.2 \lambda_{in}$ for each flow. Assume the link is fully utilized and both flows have equal delivered totals.

- a) Compute each flow's delivered total rate at the bottleneck [3 Marks].

Equal share of full link $\rightarrow$ each flow delivered total = $80/2 = 40$ Mbps.

- b) For each flow, how much of its delivered total is retransmissions [4 Marks]?

With $\lambda_{in'} = 1.2 \lambda_{in}$ and delivered total = 40, the retransmission portion is $0.2/1.2$ of the delivered total = $1/6$. So retransmissions per flow = $40 \times (1/6) \approx 6.667$ Mbps.

- c) Interpreting the retransmissions as capacity that does not carry new original data, how many Mbps of the 80 Mbps are used by retransmissions network-wide [3 Marks]?

Across two flows: $2 \times 6.667 \approx 13.333$ Mbps of the link is used by retransmissions.

Q6: Multiple Choice Questions [10 Marks]:

Q1. Compared to TCP + TLS over IP, QUIC's handshake design most directly aims to reduce:

- A. Round-trip propagation delay variance
- B. The number of required application-layer acknowledgements
- C. The number of steps before first application data**
- D. Head-of-line blocking during congestion avoidance

Answer: C

Q2. Which of the following most accurately explains why QUIC eliminates "head-of-line (HOL) blocking" at the application level?

- A. It uses larger congestion windows to pre-empt queue buildup.
- B. It multiplexes independent reliable streams within one connection, each with separate loss recovery.**
- C. It prioritizes UDP packets in router queues to avoid delay.
- D. It sends all streams in strict sequence using cumulative ACKs.

Answer: B

Q3. Which statement about QUIC's relationship to transport mechanisms is most accurate?

A. QUIC discards error control to avoid user-space overhead

B. QUIC reuses the conceptual principles of reliable transfer and congestion control in a new protocol above UDP

C. QUIC requires routers to mark packets for reliability

D. QUIC mandates flow control only at the IP layer

Answer: B

Q4. When consecutive packets on a QUIC connection are dropped due to transient path congestion, how is the sending behavior adjusted?

A. Loss is attributed to a specific stream and only that stream's rate is reduced

B. The connection's congestion controller reacts, pacing or reducing sending across all streams

C. Loss is ignored unless the UDP checksum fails

D. QUIC offloads loss handling to TLS records

Answer: B

Q5: The following is the forwarding table of a router:

P1: 11001000 00010111 00010** ***** → port 0

P2: 11001000 00010111 00011000 ***** → port 1

P3: 11001000 00010111 00011*** ***** → port 2

otherwise → port 3

Destination D = 11001000 00010111 00010110 10100001. Which port would be selected?

A. port 0

B. port 1

C. port 2

D. port 3

Answer: A

Q6: The following is the forwarding table of a router:

Q1: 11001000 00010111 00010** ***** → port A

Q2: 11001000 00010111 000101* ***** → port B

Q3: 11001000 00010111 0001011 ***** → port C

otherwise → port D

Destination D = 11001000 00010111 00010110 00000000. Which port would be selected?

- A. port A
- B. port B
- C. port C**
- D. port D

Answer: C

Q7: Which statement is accurate about two common control-plane approaches?

- A. Distributed control computes forwarding centrally; software-defined control computes routes locally on routers.
- B. Distributed control runs routing algorithms on each router; software-defined control relies on a remote controller to compute and install forwarding state.**
- C. Distributed control requires a central server; software-defined control eliminates any controller process.
- D. Both exclusively compute forwarding in input-port hardware.

Answer: B

Q8. Which statement correctly describes their roles?

- A. The data plane computes end-to-end routes; the control plane executes match-action at line rate.
- B. The data plane performs per-packet forwarding using existing tables; the control plane determines paths and installs forwarding state.**
- C. The control plane serializes bits on the physical medium; the data plane runs routing protocols.
- D. The data plane enforces admission for guaranteed QoS in a best-effort network.

Answer: B

Q9. Consider two designs for installing forwarding state:

- Design X: distributed control plane, where each router computes its own forwarding state based on local and exchanged information.
- Design Y: logically centralized controller that computes and installs forwarding entries in all routers.

Which statement best characterizes the trade-off?

- A. X always converges faster than Y because it lacks a controller bottleneck
- B. Y can react faster to global topology changes when the controller remains reachable and has accurate global state**
- C. X reduces control traffic compared to Y because routers do not exchange routes
- D. Y is strictly more stable than X regardless of network partitioning

Answer: B

Q10. What is the stated goal for input-port processing relative to the link?

- A. Exceed link speed by buffering datagrams indefinitely.
- B. Complete input-port processing at line speed.**
- C. Delay packets to reduce burstiness.
- D. Push processing into the control plane for consistency.

Answer: B